ÇANKAYA UNIVERSITY

Software Engineering Department

SENG 491 – 492 Graduation Project

Literature / Market Research

AI Driven Pet Disease-Diagnosis-Treatment Manager Tool for Veterinarians and Pet Owners (Pet Care SaaS Platform)

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1 Introduction

1.1 Purpose

This document presents the market research and existing literature review for the AI-driven Pet Disease-Diagnosis-Treatment Manager Tool. The purpose is to analyze relevant datasets, tools, and existing platforms to identify market gaps that the proposed solution will address.

1.2 Scope of this document

The document reviews the relevant market and technology landscape, covering datasets and solutions related to pet health management, AI diagnostics, and veterinary care platforms.

1.3 Definitions

Provide a table of definitions, terms, and abbreviations used in this document

Term	Description
SENG	Software Engineering
AI	Artificial Intelligence
SaaS	Software as a Service
CNN	Convolitional Neural Network
NLP	Natural Language Processing

1.4 Version History

Provide a table of table of changes to this document

Version No	**Description of change**	**Date**
1.0	Initial Release	18.10.2024

2 Proposed Project

2.1 Description

The proposed project is a **Pet Care SaaS platform** designed to provide pet owners and veterinarians with an AI-driven tool for disease diagnosis and treatment management. It allows pet owners to submit photos or videos of their pets, along with AI-based voice surveys, to receive a preliminary diagnosis. The system also facilitates price comparisons for treatments, real-time veterinarian notifications, appointment booking, and pet-friendly services like transportation.

2.2 The Need

Existing veterinary platforms do not provide AI-driven pre-diagnosis based on media, nor do they offer integration with additional services like

pet taxis or real-time health data comparison. This platform will fill the gap, providing convenience and efficiency for both pet owners and veterinarians.

2.3 Technology

The solution will be built using **React Native** for mobile development and **.NET** for the web platform, with **Python** for AI/ML models. Key technologies include:

- **TensorFlow/PyTorch** for AI-driven media-based diagnosis.
- **Google Maps API** for geolocation services (e.g., finding nearby veterinarians).
- **Stripe/PayPal APIs** for payment processing.
- **SignalR** for real-time communication between pet owners and veterinarians.
- **SQL Server/Firebase Realtime Database** for data storage.
- **Microsoft Azure/AWS** for cloud infrastructure.
- **JWT/OAuth2** for authentication and security.

2.4 Risks

- **AI Misdiagnosis**: The AI model's preliminary diagnosis might be incorrect. The platform emphasizes that veterinarians provide the final diagnosis, reducing legal and trust risks.
- **Platform Compatibility**: Maintaining consistent performance across web and mobile platforms requires rigorous testing and possibly additional resources to ensure compatibility.
- **Real-Time Synchronization**: Ensuring low latency and accurate real-time updates, especially during notifications and appointment management, is crucial to providing a seamless experience.
- **Security**: Preventing data breaches through encryption, regular audits, and security best practices will be critical in handling sensitive medical and payment information.
- **Data Privacy**: Handling sensitive health data (photos, medical surveys) securely is essential, requiring encryption, regulatory compliance, and vulnerability testing.

3 Existing/Similar Solutions

3.1 Product 1: Vet-U

- **Link**: GitHub Repository: Vet-U
- **Solution Description**: A veterinary application focusing on health management, including appointment scheduling and communication between pet owners and veterinarians.

- **Shortcomings**: Vet-U lacks AI-driven diagnostics, multimedia input for pre-diagnosis, and real-time synchronization of health data with veterinarians.

3.2 Product 2: VetHeal

- **Link**: GitHub Repository: VetHeal App
- **Solution Description**: A mobile app designed for managing pet health and scheduling veterinary appointments.
- **Shortcomings**: The app does not provide AI-driven diagnostics, multimedia input for health assessments, or auxiliary services like price comparisons or pet taxis.

3.3 Product 3: PetCare App

- **Link**: GitHub Repository: PetCare App
- **Solution Description**: An application offering basic pet care functionalities, such as appointment scheduling and health tracking.
- **Shortcomings**: PetCare does not offer AI diagnosis features, nor does it integrate auxiliary services such as pet transportation or price comparisons.

3.4 Product 4: TTCare App

- **Link:** https://www.ttcareforpet.com/
- **Solution Description:** Pawp has developed an AI-powered diagnostic tool designed to analyze pet health conditions directly from phone photos. Leveraging computer vision, this AI system enables pet owners to quickly assess potential health issues by uploading images to the platform, which then provides preliminary diagnostic suggestions. The tool is intended to make basic health assessments more accessible, offering owners insights they can discuss with veterinarians for further confirmation and treatment.
- **Shortcomings:** This tool focuses solely on photo analysis and does not incorporate other multimedia inputs like video or additional health metrics. Furthermore, the preliminary diagnoses are limited to visual symptoms and do not support extensive health survey data from pet owners

4 Conclusion

The current market for veterinary care platforms provides essential tools for health monitoring and appointment scheduling. However, these platforms lack comprehensive AI-driven diagnostics and auxiliary service integrations, such as transportation, price comparison, and multimedia pre-diagnosis. The proposed Pet Care SaaS platform addresses these gaps by integrating a robust AI system for preliminary diagnoses, real-time health data synchronization, and a wider range of pet services into one unified solution.

5 References and Data Sources

- Animal Bite Dataset (Kaggle)

  A dataset on animal bites, relevant for understanding health incidents involving pets.

- Animal Classification Dataset (GTS.AI)

  A dataset for classifying various animal species, which is essential for pet disease diagnostics.

- Animal Disease Classification (Kaggle)

  A dataset on animal diseases, crucial for training AI models in the proposed system.

- Animal Detection (Kaggle)

  Focuses on animal detection from images, valuable for the media-based pre-diagnosis feature.

- Avacta Animal Health Dataset (Kaggle)

  A comprehensive dataset related to animal health and treatment efficacy, useful for improving AI models.

- Veterinary Science Resources (University of Pretoria)

  Offers access to veterinary science literature, supporting the scientific basis of AI diagnoses.

- Disease Prediction using Machine Learning (Kaggle)

  A dataset on predicting diseases with machine learning, relevant for building predictive models.

- Dog Skin Disease Dataset (Kaggle)

  A dataset focusing on skin diseases in dogs, which can improve the accuracy of AI diagnoses.

- Artificial Intelligence and its Application in Animal Disease Diagnosis (Research Paper)

  A research paper exploring the use of AI in animal disease diagnosis, providing insights for developing the platform’s AI component.

[Software Engineeering Department, "Graduation Projects," SENG, [Online].] Available: https://seng.cankaya.edu.tr/graduation-projects/. [Accessed 28 June 2024].